Assignment 1 Solution

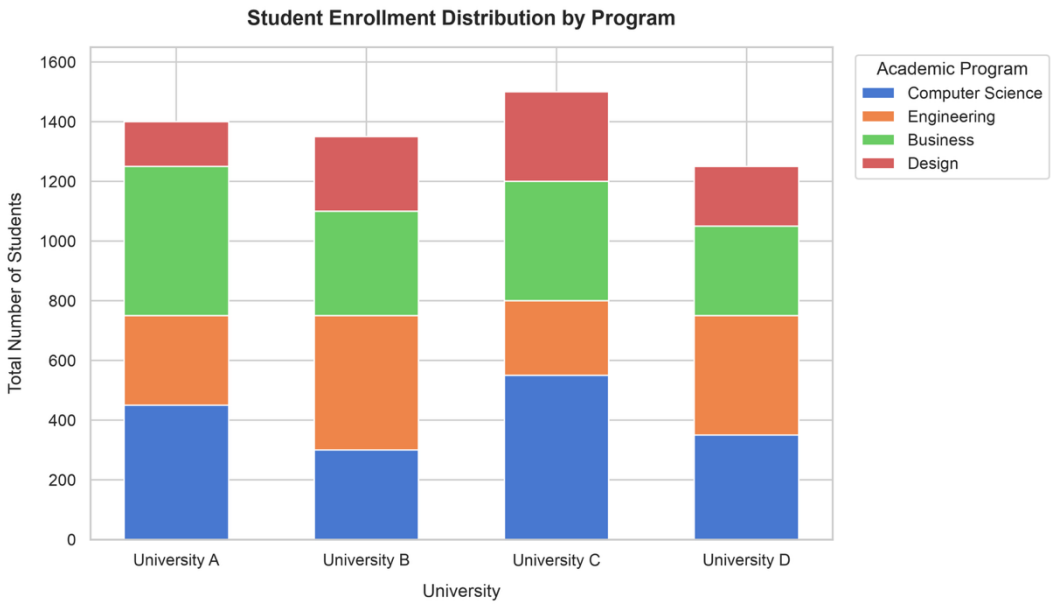
4DV805 / 4DT909 - Information Visualization

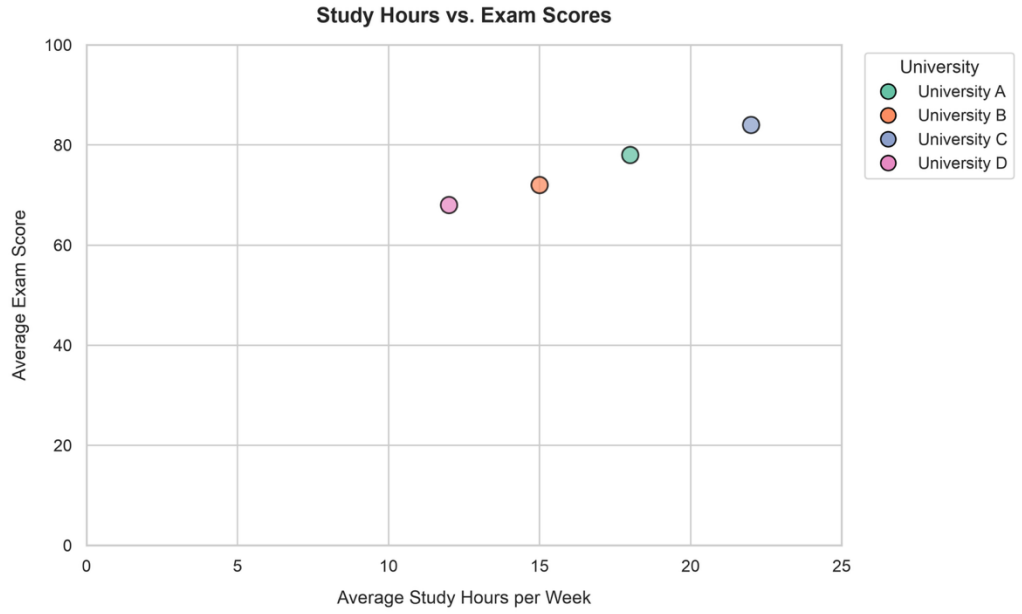
30 September 2026

Part 1

You are given **2** charts below. Your task is to analyze each visualization. For each chart, provide the following:

- **Data Attributes and Types:** Identify all data attributes represented in the visualization and specify the data type of each attribute
- **Visual Encodings:** Identify the low-level visual encodings used to represent each data attribute.

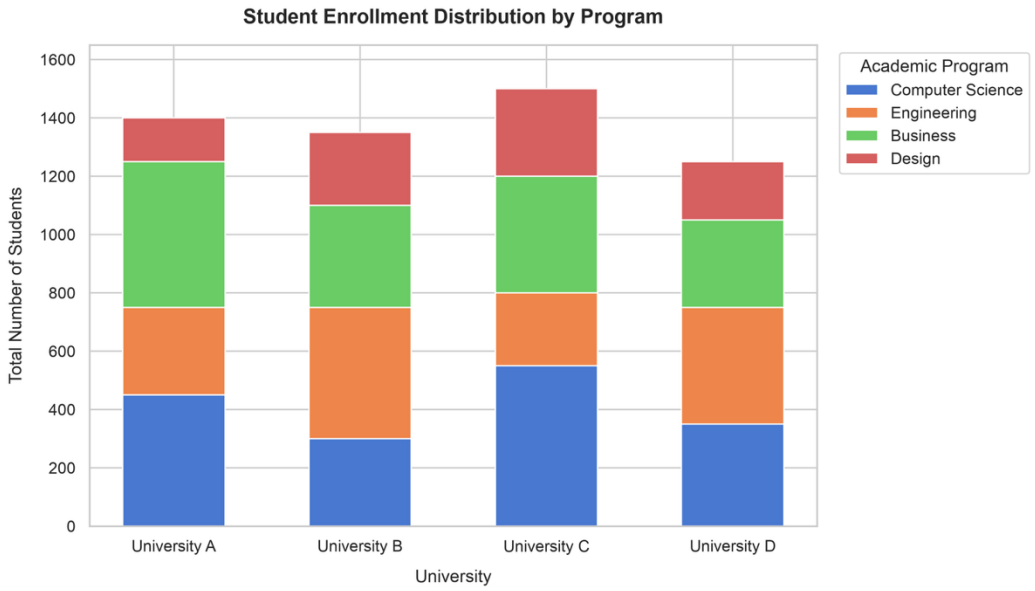




2.5.1 Preattentive Features

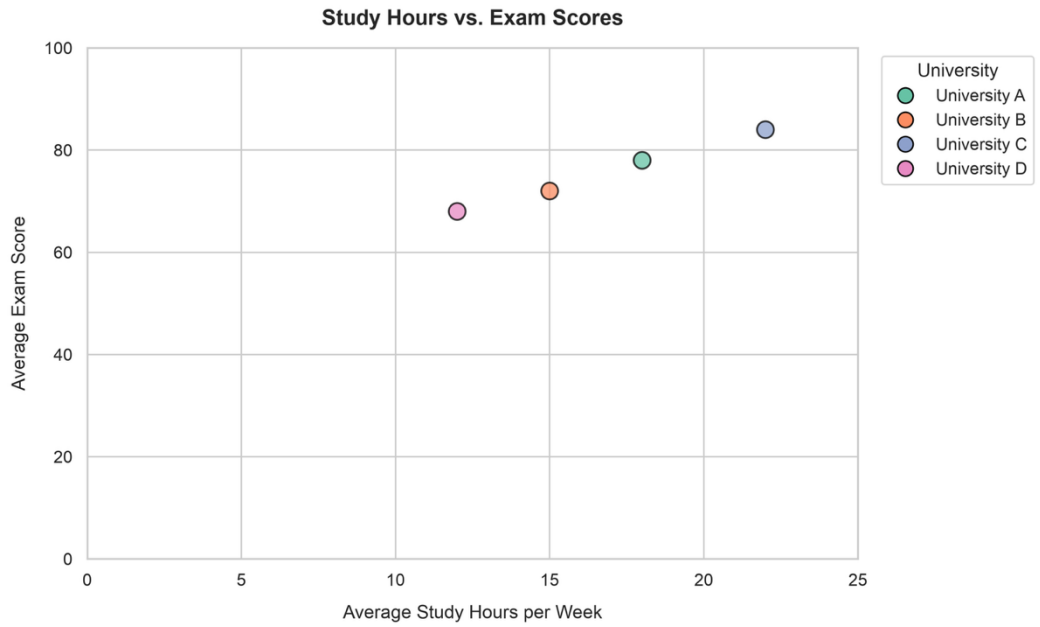
- Form
  - Line orientation
  - Line length
  - Line width
  - Size
  - Curvature
  - Spatial grouping
  - Added marks
  - Blur
- Color
  - Hue
  - Intensity
- Motion
  - Flicker
  - Direction of movement
- Spatial Position
  - 2D position
  - Convex/concave shape from shading

Chart 1



Attribute	Data Type	Encoding
University	Categorical / Nominal	Position
Number of Students	Numerical / Quantitative	Length / Area / Size
Academic Program	Categorical / Nominal	Color

Chart 1



Attribute	Data Type	Encoding
University	Categorical / Nominal	Color
Avg. Study Hours per Week	Numerical / Quantitative	Position (X)
Avg. Exam Score	Categorical / Nominal	Position (Y)

Part 2

Using the data attributes identified in Part 1, **design a new visualization** that represents or communicates all of the attributes within a **single chart**.

Your submission should:

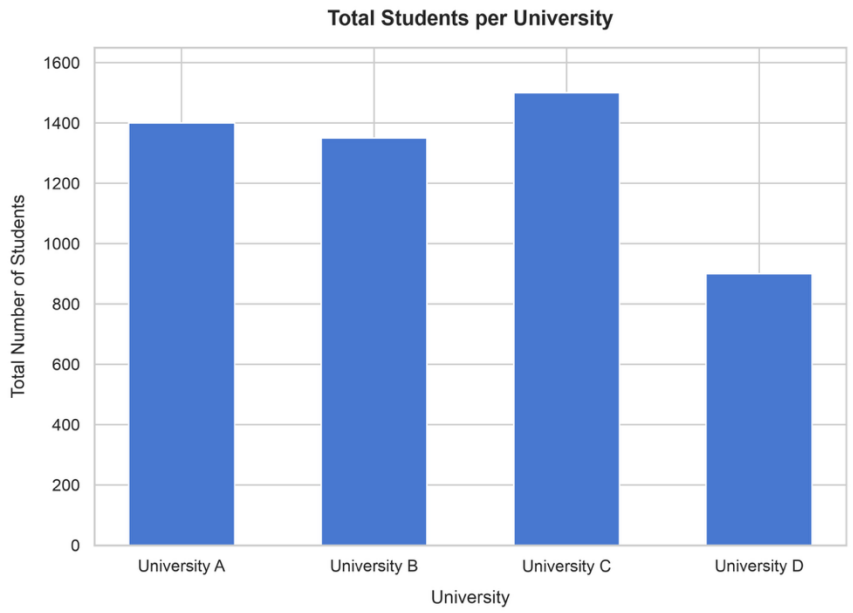
- Identify the base chart type or visualization technique you have selected.
- Specify how each data attribute is visually encoded.
- Explain the rationale behind each encoding choice.
- Explain how your design addresses any limitations or problems identified in Part 1.
- Ensure that the resulting visualization is interpretable and allows the viewer to effectively understand the data.

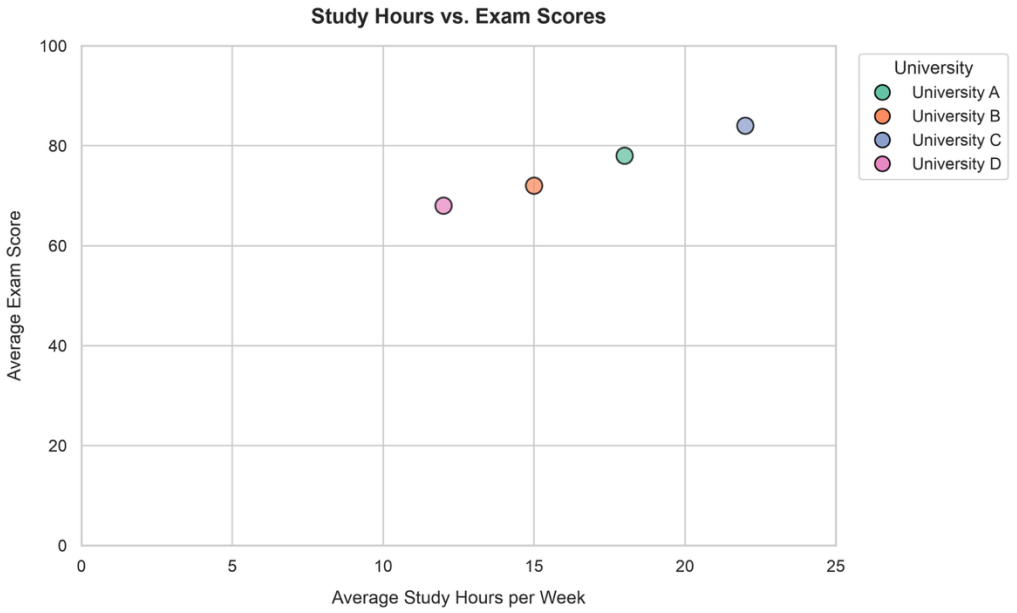
Your proposed visualization may be an adaptation or extension of an existing, widely used chart type.

Design Hints:

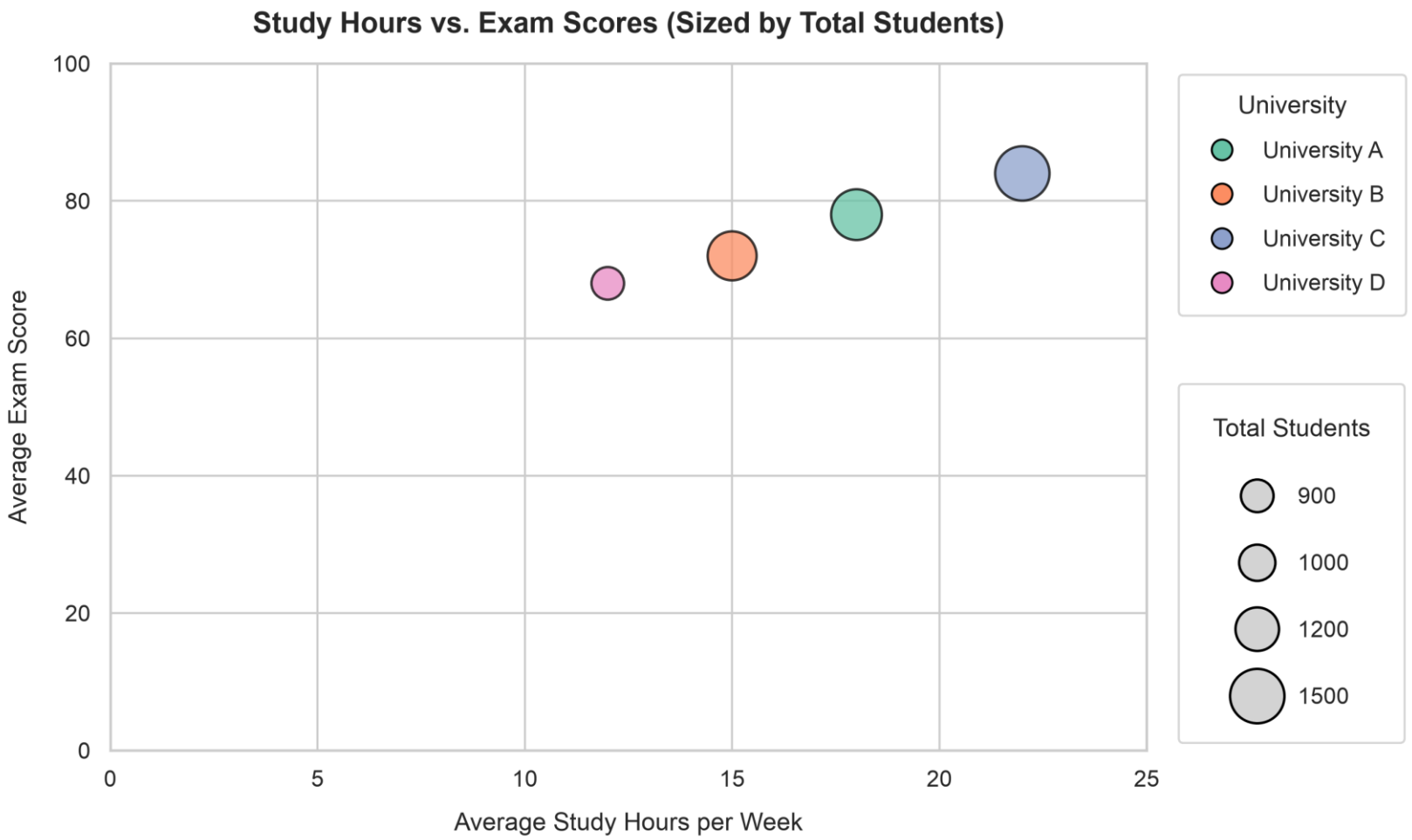
- How many data attributes must be represented?
- What visual encodings are available to represent those attributes?
- Which existing visualization technique can accommodate or be modified to accommodate the required number of encodings while remaining interpretable?

Part 2: Simplified

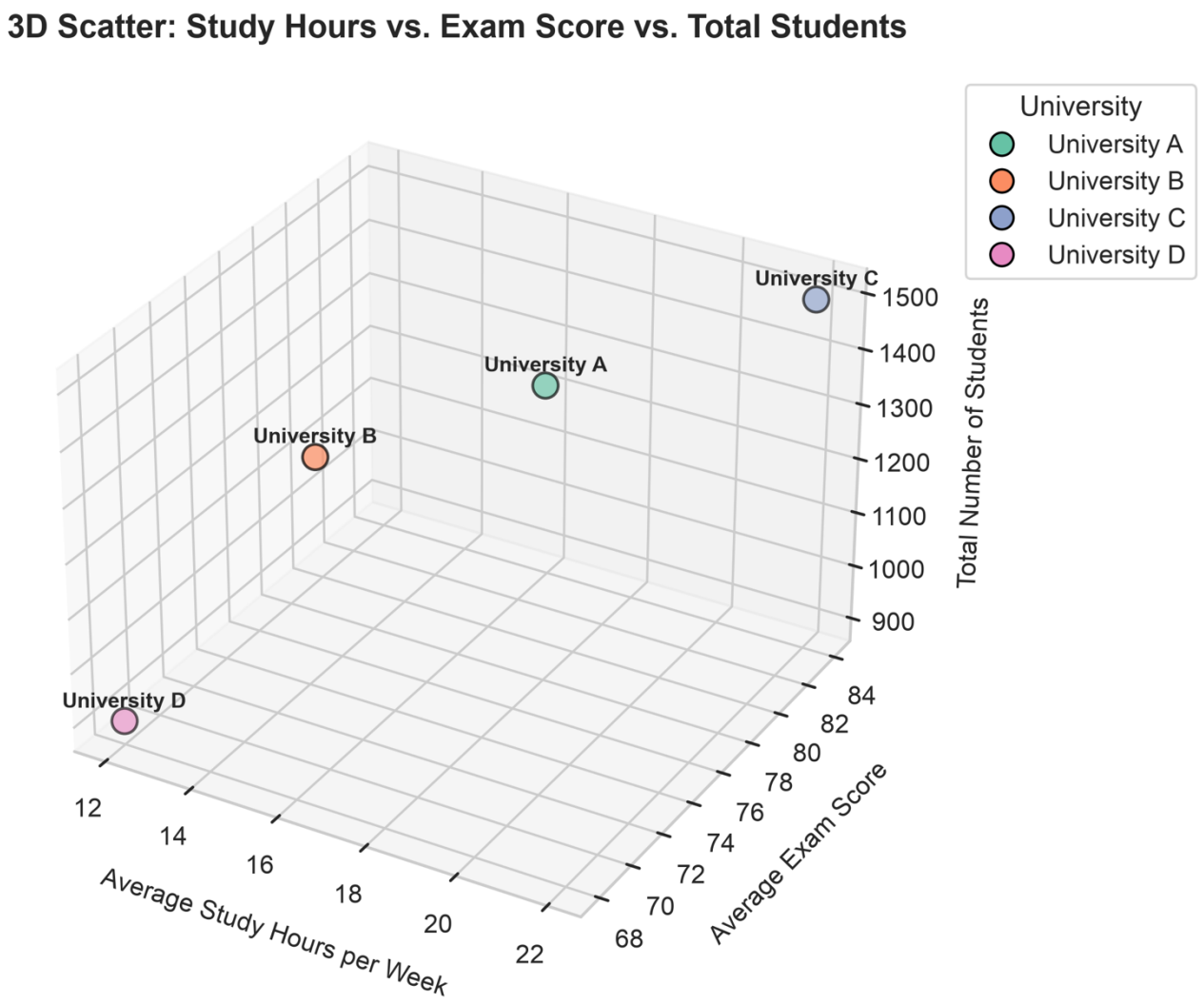




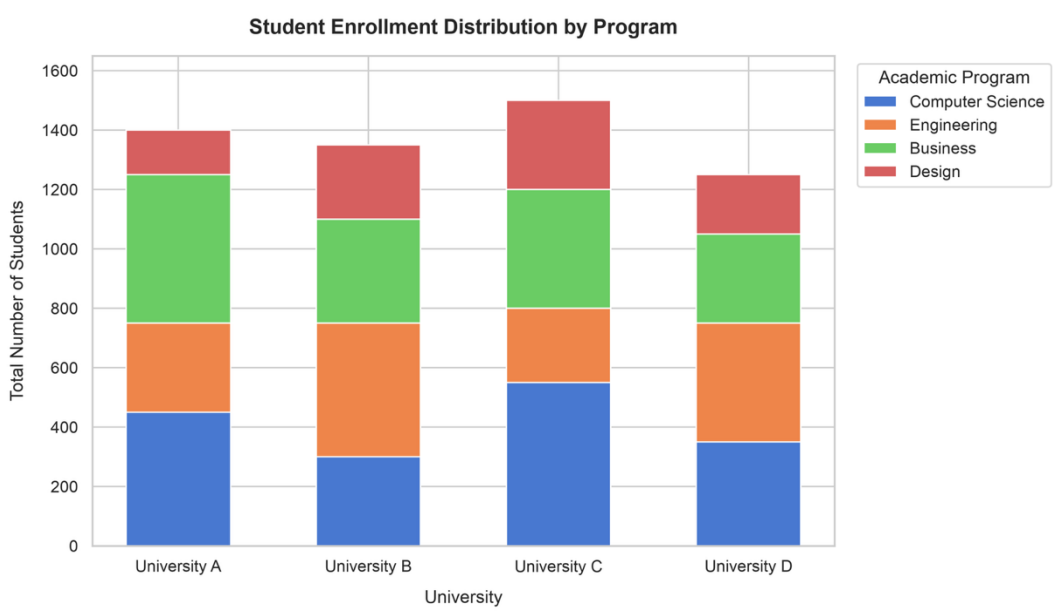
Bubble Plot

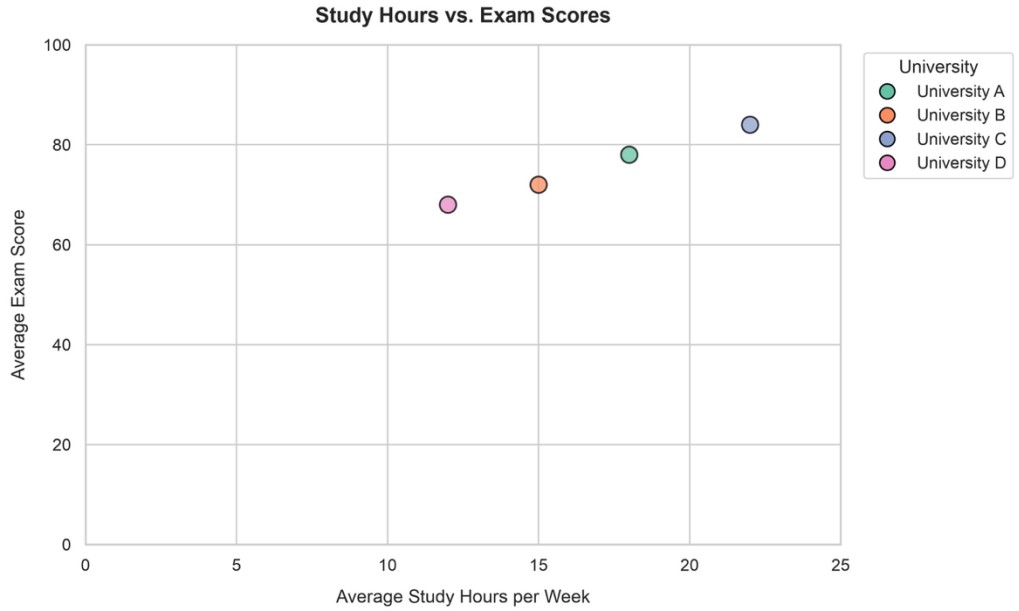

Study Hours vs. Exam Scores (Sized by Total Students)
Average Exam Score
Average Study Hours per Week
0
20
40
60
80
100
0
5
10
15
20
25
University
University A
University B
University C
University D
Total Students
900
1000
1200
1500

3D Scatterplot

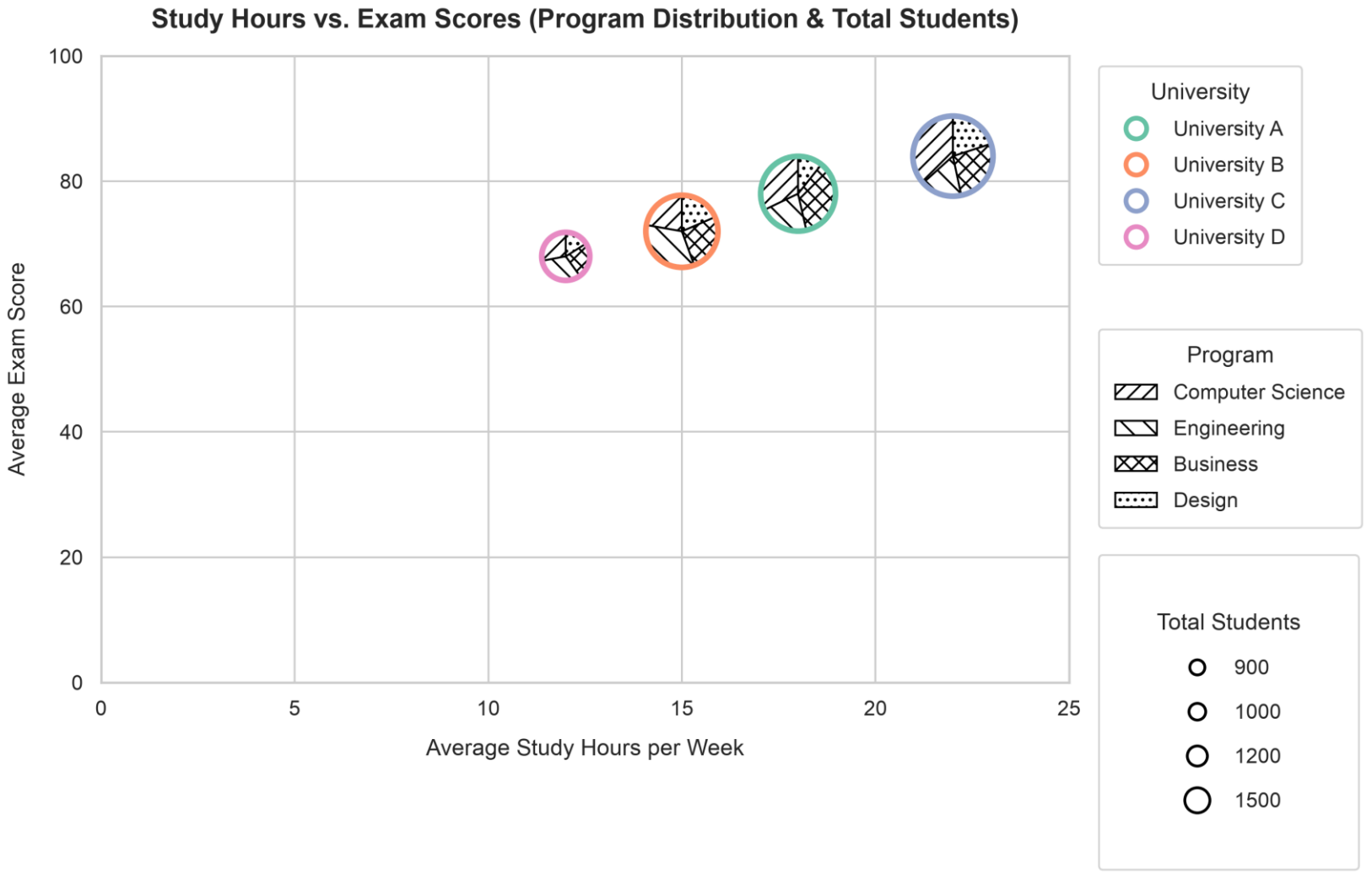


Part 2: Complex

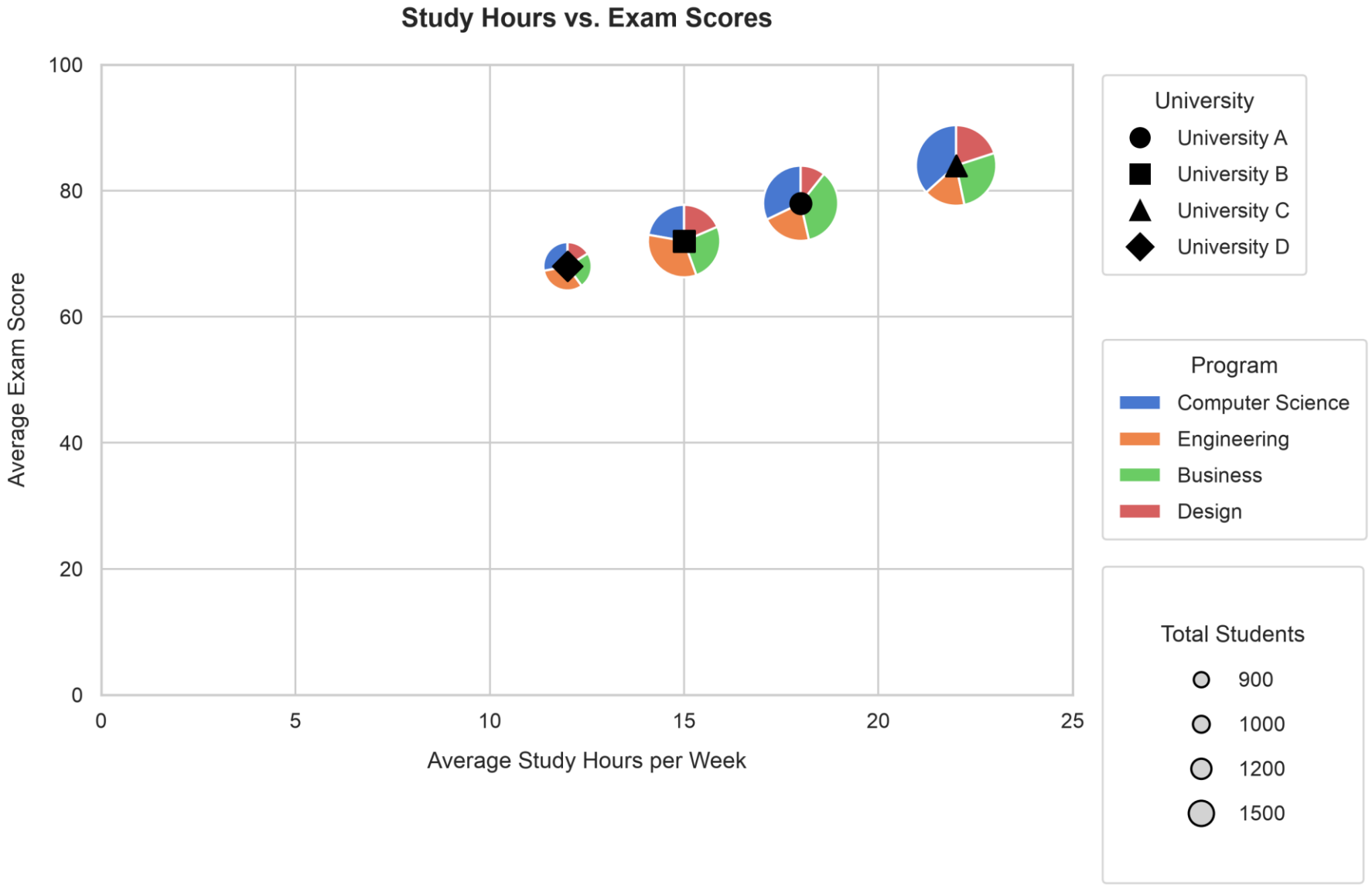




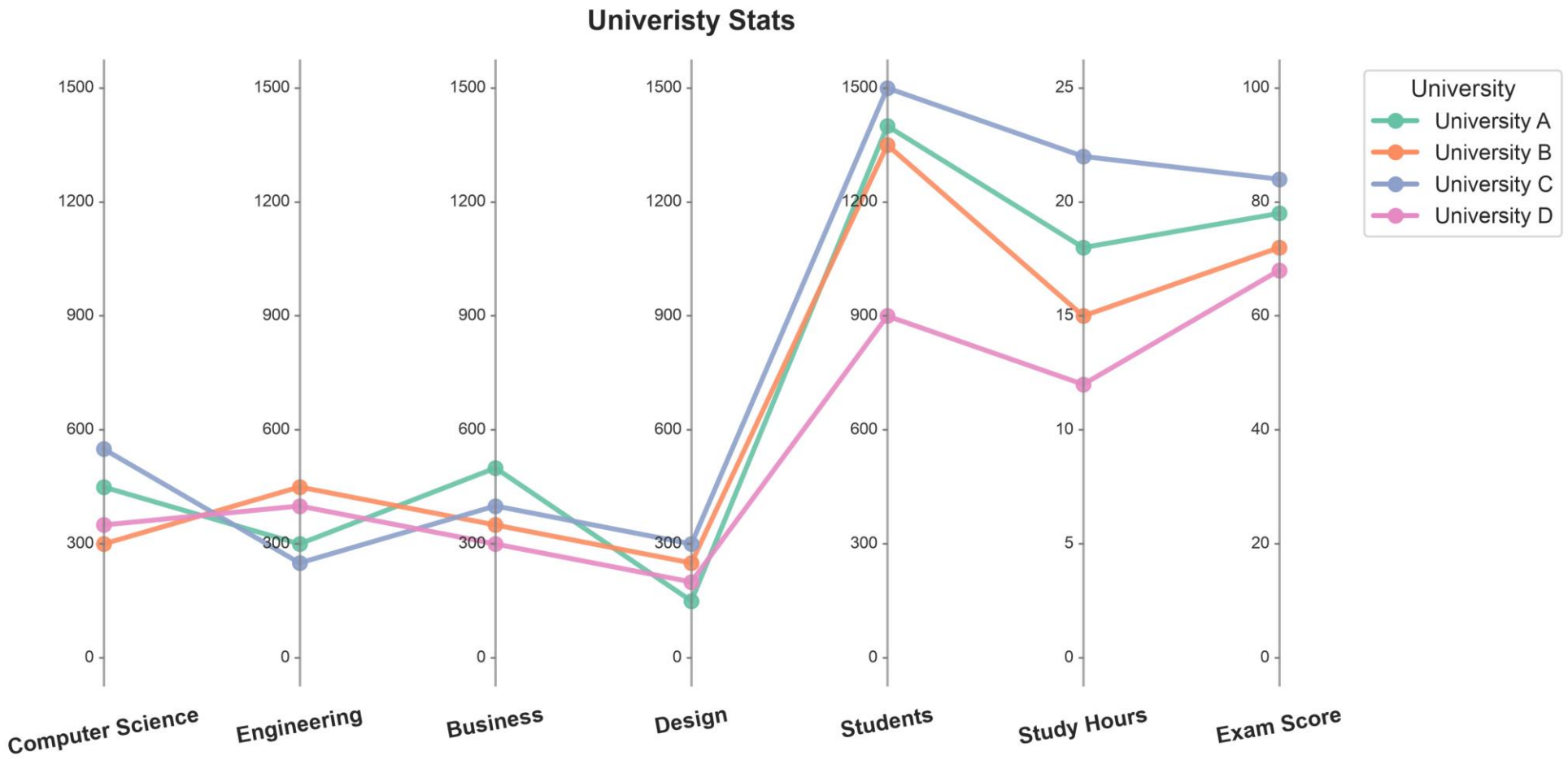
Scatterplot with Complex Visual Representation for Points



Scatterplot with Complex Visual Representation for Points



Parallel Coordinates



Marks

A mark is a basic graphical element in an image. Marks are geometric primitive objects classified according to the number of spatial dimensions they require.

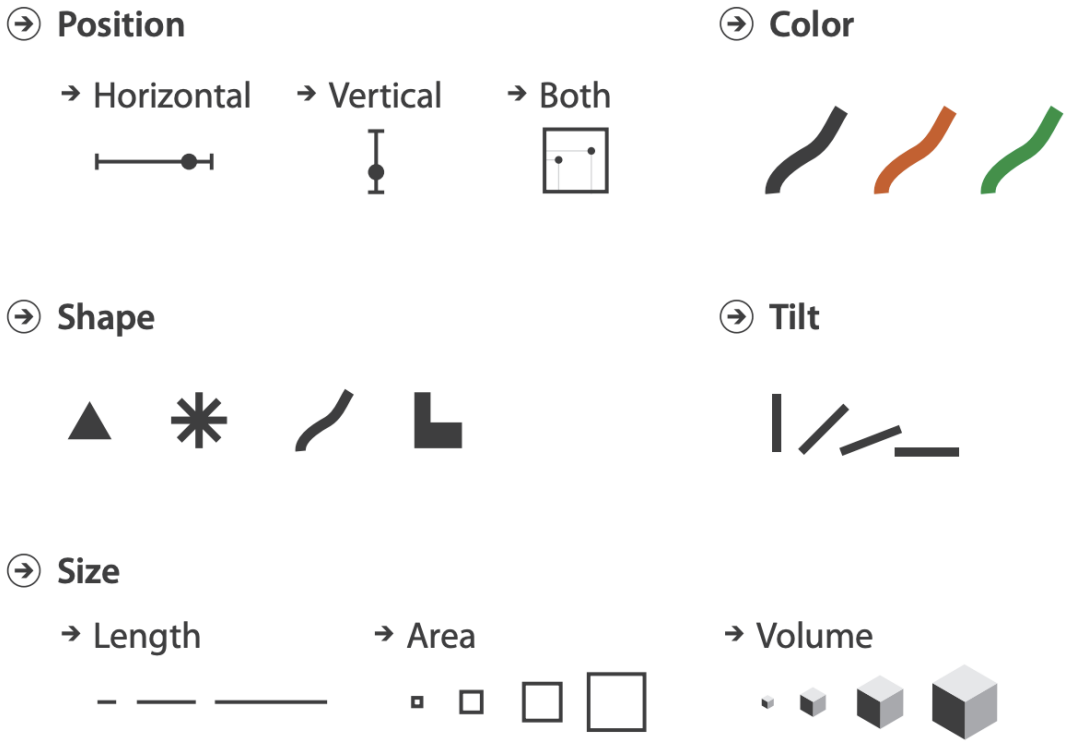


Visualization analysis and Design
by Tamara Munzner

Published: 2015
ISBN: 9781466508910

Visual Channels

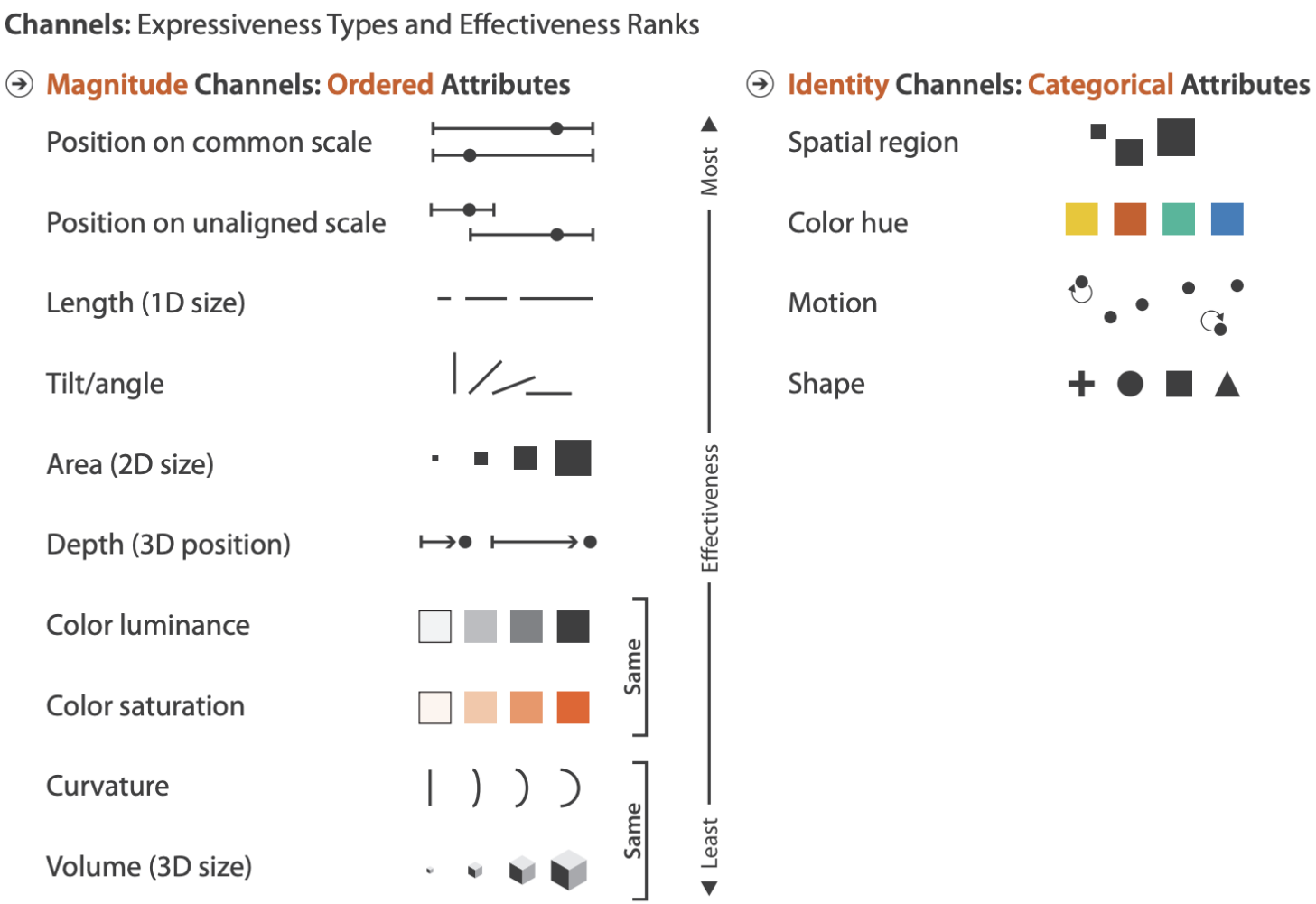
A visual channel is a way to control the appearance of marks.



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Preattentive Feature / Visual Encodings / Visual Channels

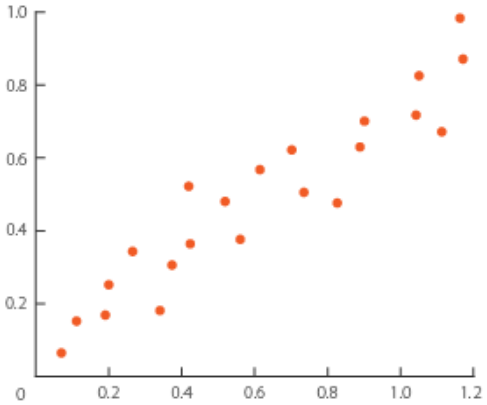


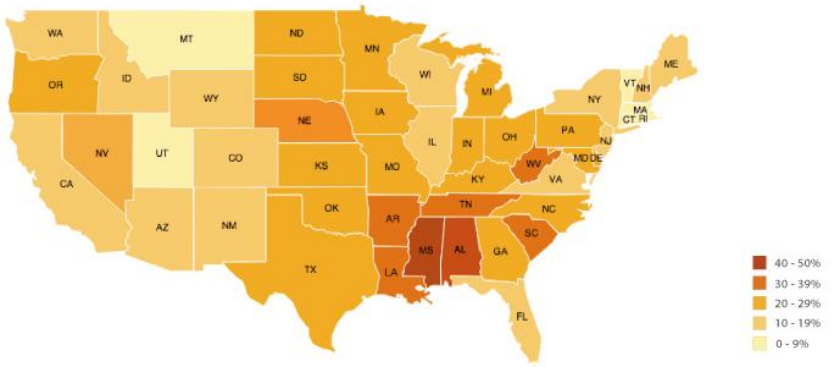
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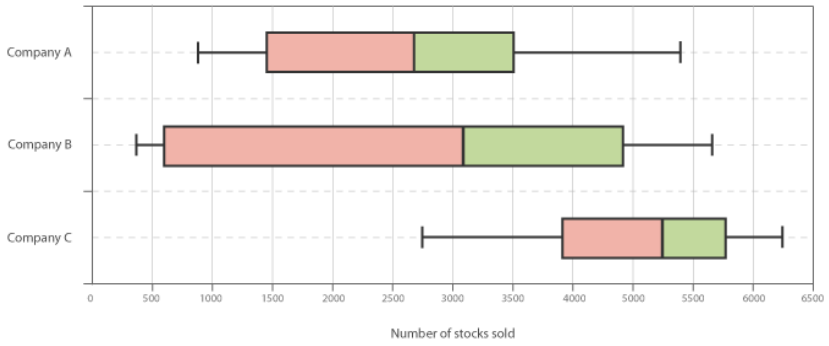
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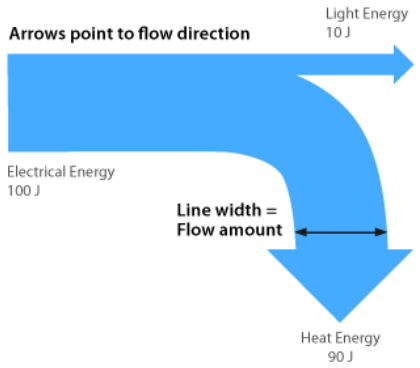
Preattentive Feature: Position

Where visual marks are placed and their relative position to each other communicate information.







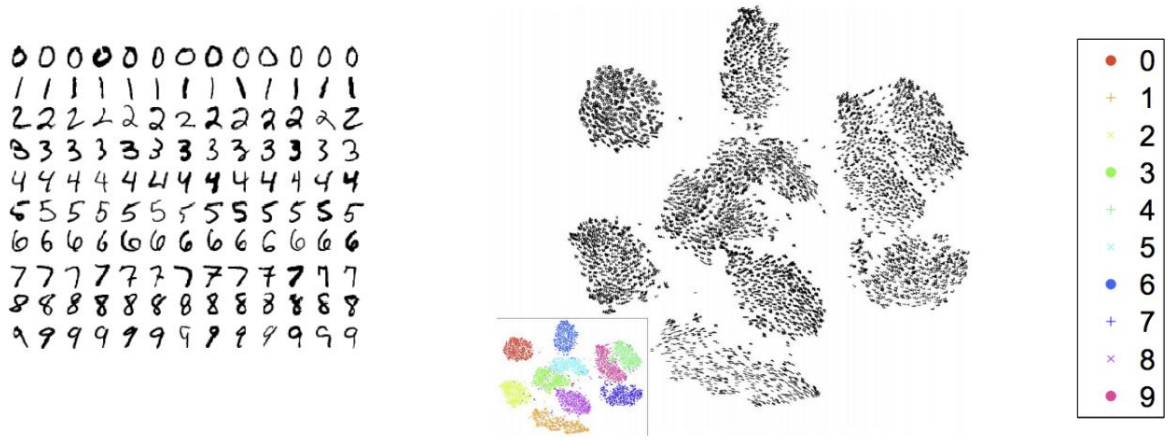


Preattentive Feature: Position

Where visual marks are placed and their relative position to each other communicate information.

Dimensionality Reduction

- T-SNE (van der Maaten & Hinton, 2008)



http://projector.tensorflow.org